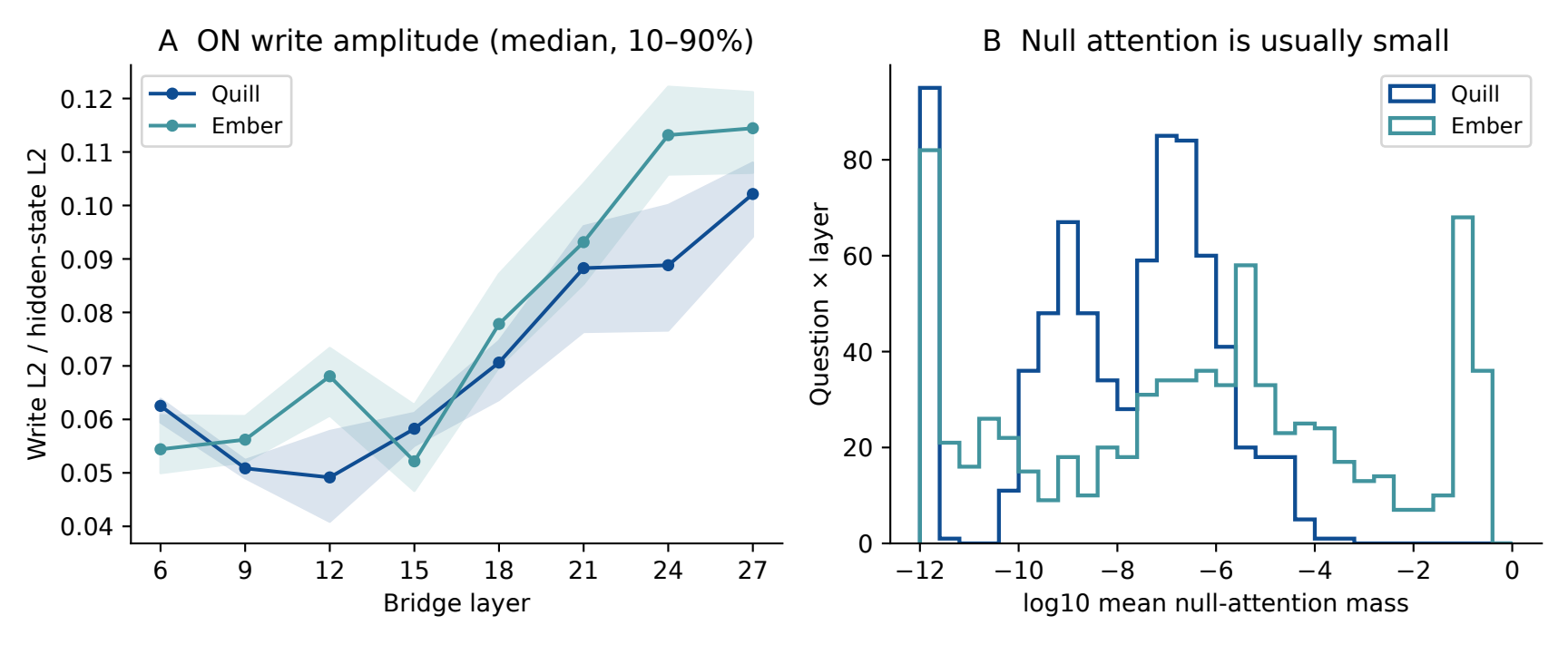
A ON write amplitude (median, 10–90%)
Quill
Ember
Write L2 / hidden-state L2
0.12
0.11
0.10
0.09
0.08
0.07
0.06
0.05
0.04
6
9
12
15
18
21
24
27
Bridge layer
B Null attention is usually small
Quill
Ember
Question × layer
80
60
40
20
0
−12
−10
−8
−6
−4
−2
0
log10 mean null-attention mass